

Luisa Cicolini

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🆔 0009-0007-9831-6968 | 📍 Cambridge, UK

creativity and formal methods enthusiast // PhD student

Projects

Society

Formalized Semantics For Hardware Verification

Location

September 2024 - present

I am interested in bridging maths and hardware verification, combining bitvectors and coinduction for the verification of latency-insensitive hardware abstractions.

Society

Formalized RISC-V Instruction Set Architecture

Location

March 2025 - present

I work on a fully-bitblastable library for the RISC-V Instruction Set Architecture, proven equivalent to the Sail processor model and aimed at verifying LLVM’s backend, particularly the code generation pass.

Society

Lean4 BitVec

Location

July 2024 - present

I have contributed to Lean4, working on the development of Lean’s bitvector library and verified bitblaster.

Society

Model Checking For CIRCT

Location

September 2023 - March 2026

I worked on formalizing the semantics of CIRCT’s Finite State Machine (FSM) dialect, enabling CIRCT to integrate verification at a higher level of the stack.

Talks & Publications

- L. CICALINI, S. L. Kuhn, O. Yasar, “Scaling Certified Instruction Selection for LLVM IR Through Bitblasting,” EuroLLVM’ 2026
- H. Böving, S. B. Mala, L. CICALINI, et al., “Interactive Bitvector Reasoning using Verified Bit-Blasting,” OOPSLA’ 25
- B. Healy, L. CICALINI, “Exploiting MLIR Abstractions for Hardware Verification,” LLVM-dev 2024
- L. CICALINI, F. Carloni, et al., “One Automaton to Rule Them All…” , CGO’ 24

Education

Society

University of Cambridge, UK

Location

Date

- 2025-present, Ph.D. Student with the Cascade Centre
- 2024-2025, Research Assistant

Society

Politecnico di Milano, Italy

Location

Date

- 2021-2023, M.Sc. Computer Science and Engineering
- 2017-2020, B.Sc. Energy Engineering

Certificates

- Goethe Zertifikat B2 - 2016
- Cambridge Proficiency English - C2 - 2017
- TOEFL iBT - 116/120 - 2023

Misc

- Volunteered with kids and elderlies.
- Piano & singing enthusiast.
- Student volunteer at DATE, PLDI, OOPSLA.